

AIDAN O'BRIEN

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EDUCATION

University of Wisconsin - Madison

Sep 2023 - May 2027

B.S. Computer Science

GPA: 3.9/4.00

Relevant Coursework: Algorithms, Operating Systems, Big Data Systems, Artificial Intelligence & Discrete Math

EXPERIENCE

Software Engineer Intern

May 2026 - August 2026

Netflix

Los Gatos, CA

- Drove **\$15M+** in projected savings by leading a UX-focused A/B test on the member cancel flow to improve subscriber save rate, spanning the React frontend to several upstream GraphQL and Falcor endpoints.
- Architected a cross-repo migration unifying Change Plan routing across 3 siloed routing paths from hardcoded logic to Netflix's Odyssey state-machine platform, migrating the first path end-to-end in my final week.
- Pioneered monitoring-as-code for my team's backend services, building a Gradle mechanism that publishes dashboards/alerts as version-controlled JSON via existing Jenkins CI, replacing manual UI configuration.
- Patched a critical regex bug on cssjanus, an **open-source** RTL CSS library with **600,000+** weekly downloads.

Founding Engineer

Dec 2025 - May 2026

Intelligible AI

Remote

- Built the data ingestion pipelines for the first **4** database connectors (Postgres, MySQL, Snowflake, BigQuery) across multiple Lambdas + Step Functions, enabling our customers to securely integrate their production data.
- Integrated Fivetran Connect Card with a webhook-driven sync lifecycle across **8** Lambdas, expanding the platform from **4** native connectors to **700+** supported data sources without per-source auth code.
- Led migration of infrastructure-as-code from AWS CDK to SST, cutting PR deploy times by **85%** and enabling local preview environments that accelerated the team's ship cadence from weekly to daily.

Software Engineer Intern

June 2025 - Aug 2025

CargoLabs

Remote

- Eliminated **8+ hours/month** of manual work by automating bordereau reporting w/ an AWS + SQL cron job.
- Standardized authentication pipelines across **7 AWS Cognito** user pools while implementing consistent UI design, creating unified user workflows and reducing authentication complexity across all portals.
- Built a comprehensive customer service portal enabling the CargoLabs team to efficiently manage and troubleshoot client website problems, reducing average issue resolution time by **45-60%**.

PERSONAL PROJECTS

BadgerBase 🐾 | *Next.js, Hono, Bun, PostgreSQL, Redis, MCP, Railway*

- Created a web app that has reached **2000+ active student users** that optimizes the UW Madison course sign-up experience by integrating course, professor, and grade data from multiple sources.
- Shipped an OAuth-secured **MCP server** that lets AI assistants like Claude search the catalog, pull grade & professor data, and read a student's enrollment alerts through scoped tools.
- Designed a **PostgreSQL** database serving enrollment & grade data for **20,000+** sections through a Hono/Bun + Kysely API, with typo-tolerant pg_trgm fuzzy search and a Redis caching layer.
- Built a self-hosted auth (better-auth) and email stack on a Mox mail server that alerts students when competitive sections open, running at **\$22/year** versus **\$240/year** for third-party email providers.

Local LLM Inference Lab | *llama.cpp, vLLM, Python, asyncio, Apple Metal*

- Built an instrumented inference lab on an M4 Pro (24GB) serving quantized models to **8** concurrent coding agents via llama.cpp, with an asyncio load generator logging per-request prefill/decode timings.
- Modeled KV-cache and memory-bandwidth limits from first principles to predict decode throughput and max agent concurrency before each experiment, landing within **20%** of measured results.
- Cut time-to-first-token by **60×** with slot-pinned prefix caching and **doubled** agent capacity via 8-bit KV-cache quantization, then benchmarked llama.cpp against vLLM's PagedAttention on the same workload.

SKILLS

Languages: Python, TypeScript/JavaScript, Java, C, SQL

Frontend: Next.js, React, Tailwind, Zustand, PostHog, Vercel

Backend: Node.js, Bun, Hono, GraphQL, Spring Boot, Flask, DynamoDB, PostgreSQL, Redis, MySQL, Supabase

Infrastructure: AWS (Lambda, Step Functions, ECS, S3), SST, Docker, Railway, GitHub Actions